Predictive and Slice-Aware Load Balancing and Auto-Scaling for 5G Core AMFs

Addressing the 40s Container Cold-Start Lag and URLLC QoS Violations

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Motivation: 5G Core AMF Bottleneck

Role of the AMF (Access and Mobility Management Function):

- Terminates N1 (NAS signaling with UEs) and N2 (NG-AP with gNodeBs).
- Handles subscriber registration, authentication, mobility, and PDU session setup.
- Operates as a cloud-native microservice inside Docker containers in 5G SBA.

The Heterogeneous Slice Challenge:

- **URLLC:** Ultra-Reliable Low-Latency (< 5.0 ms SLA limit).
- **eMBB:** High-bandwidth streaming (10–25 ms target).
- **mMTC:** Millions of IoT sensors with massive burst potential.

The Problem in Practice

When millions of IoT devices reboot simultaneously (e.g., smart power meters after an outage), a **signaling storm** strikes the AMF.

- Control-plane CPU saturates rapidly.
- Incoming queues explode.
- Mission-critical URLLC signaling is trapped behind delay-tolerant IoT messages, causing massive SLA violations!

Literature Survey & Research Gap

ScaleLB Baseline (IEEE ICOIN 2025):

- Introduced an elastic load balancer in front of AMF pools using gNB-to-AMF ratio thresholds:

$$\text{Proactive: } \frac{B}{A \cdot \alpha} \geq \gamma = 0.80 \qquad \text{Reactive: } \frac{B}{A \cdot \alpha} > 1.0$$

Vulnerability 1: 40s Cold-Start Lag

Container spin-up requires $T_{\text{boot}} \approx 40\,\text{s}$. Because ScaleLB only triggers scaling *after* load crosses the threshold, queues build up during the 40s lag, causing latency spikes ($> 330\,\text{ms}$).

Vulnerability 2: Slice-Agnostic Single FIFO

ScaleLB uses a single shared FIFO queue. During an IoT burst, URLLC signaling suffers the same queuing delay as mMTC, severely violating the 3GPP 5.0 ms SLA.

Problem Statement & Core Objectives

Formal Problem Statement

Design an ingress control-plane load balancing and auto-scaling mechanism that strictly guarantees sub-5 ms latency for URLLC slices and eliminates the 40-second container startup backlog under arbitrary 5G IoT traffic surges.

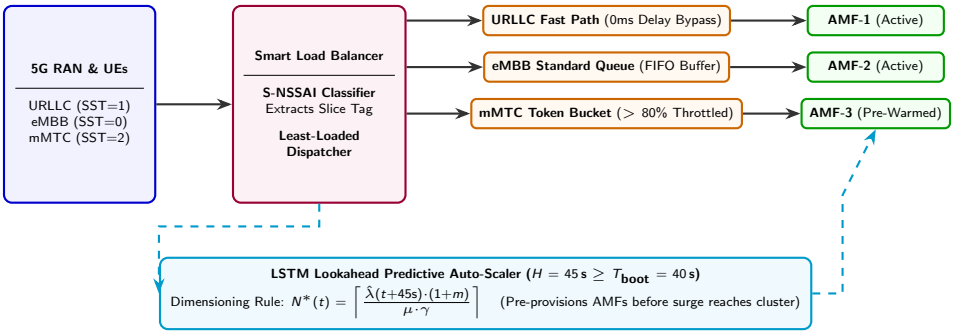
Objective 1 (O1): Slice-Aware Priority Queuing

- Parse 3GPP S-NSSAI slice headers at ingress.
- Grant non-blocking zero-queue bypass for URLLC (< 5 ms).
- Token-bucket rate limit mMTC when cluster CPU $> 80\%$.
- Starvation-free aging for delay-tolerant packets.

Objective 2 (O2): LSTM Predictive Scaling

- Maintain 60-second sliding history window.
- Lookahead horizon $H = 45\,\text{s} \geq T_{\text{boot}} = 40\,\text{s}$.
- Predict surge before it reaches the core.
- Pre-warm AMF containers to absorb the wave at $t = 0$.

System Architecture Overview



Slice-Aware Priority Queuing Formulation

1. URLLC Fast Path (SST = 1):

- Zero-queue bypass: skips ingress buffers entirely.
- Directly dispatched to lowest-loaded active AMF:

$$\text{Target AMF} = \arg \min_{i \in \mathcal{A}_{\text{active}}} \text{CPU}_i(t)$$

- 10–15% headroom reserved on all active AMFs.

2. eMBB Standard Queue (SST = 0):

- Standard FIFO queue for normal mobile traffic.
- Typical latency maintained at 10–18 ms.

3. mMTC Token Bucket with Aging (SST = 2):

- When cluster utilization $> 80\%$, token bucket limits mMTC release rate:

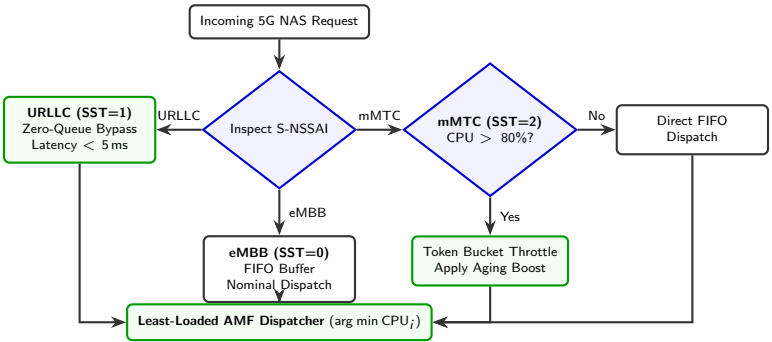
$$R_{\text{mMTC}}(t) = \max\Big(0.10 \cdot C(t),\ C(t) \cdot (1 - \text{CPU}(t))\Big)$$

- **Anti-Starvation Aging**: Buffered packets earn priority over time:

$$P_i(t) = P_{\text{base}} + \omega_{\text{age}} \cdot (t - t_{\text{arrive},i})$$

- When $P_i(t)$ crosses eMBB threshold, the request is promoted to avoid dropped sessions.

Packet Scheduling Decision Flow



LSTM Lookahead Predictive Auto-Scaling

Mathematical Formulation:

- Observation sliding window: $\mathbf{w}_t = [\lambda(t-59), \ldots, \lambda(t)] \in \mathbb{R}^{60}$
- Forecast arrival rate 45s ahead: $\hat{\lambda}(t+H) = \text{LSTM}_\theta(\mathbf{w}_t)$
- Lookahead condition: $H = 45\,\text{s} \geq T_{\text{boot}} = 40\,\text{s}$

Capacity Dimensioning Scaling Rule

$$N^*(t) = \left\lceil \frac{\hat{\lambda}(t+H) \cdot (1+m)}{\mu \cdot \gamma} \right\rceil$$

- $\mu = 100\,\text{req/s}$ (nominal AMF service rate, $T_s = 10\,\text{ms}$)
- $\gamma = 0.80$ (target CPU utilization threshold)
- $m = 0.10$ (10% safety margin against model under-prediction)

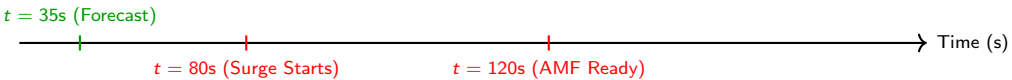
Scale-In Hysteresis Cooldown: To prevent instance thrashing, scale-in is restricted by a 30-second cooldown window and graceful connection draining.

Eliminating the 40-Second Cold-Start Lag

ScaleLB Baseline (Reactive):

40s Boot Delay (Overloaded Cluster)

Backlog $> 8,800$ **req** $\Longrightarrow$ **Latency 338ms (FAIL)**



Proposed (LSTM Lookahead $H = 45$**s):**

Pre-warming AMF **after already at 400 req/s** $\Longrightarrow$ **0s Lag, URLLC** $<$ **5ms (PASS)**

Experimental Results: Benchmark Comparison

Performance Under 320 req/s Flash Crowd Surge (ScaleLB Fig. 5 Trace):

Configuration	Cold-Start Lag (s)	URLLC p99 Latency	mMTC p99 Latency	URLLC SLA	AMF-sec (Cost)
ScaleLB Reactive [?]	54 s	338.7 ms	339.3 ms	FAIL (25.8%)	647
ScaleLB Proactive ($\gamma = 0.8$)	55 s	334.1 ms	336.5 ms	FAIL (25.8%)	649
Warm Standby (1 idle spare)	**0 s**	32.9 ms	32.7 ms	PASS	1024
O1 only: Slice Priority Queue	54 s	11.1 ms	508.6 ms	PASS	**647**
O2 only: LSTM Prediction	40 s	333.1 ms	336.5 ms	FAIL (18.3%)	807
Proposed System (O1 + O2)	**0 s**	**3.15 ms**	**499.9 ms**	PASS (0.0%)	**807**

- **ScaleLB Fails SLA**: Latency spikes to 338.7 ms under surge (> 5.0 ms limit).
- **Proposed System Guarantees SLA**: URLLC bounded at **3.15 ms** (100% compliance).
- **Resource Savings**: 21.2% lower infrastructure cost than Warm Standby (807 vs 1024 AMF-s).

Platform Implementation Visuals

Live Slices & Forecast Dashboard:



Top: Traffic demand vs. 45s LSTM lookahead vs. provisioned AMF capacity. Bottom: Per-slice latency curves.

AMF Container Fleet Scaling Topology:



Real-time visualizer showing elastic lifecycle states: `ACTIVE`, `BOOTING (40s lag)`, and `STOPPED`.

Key Takeaways & Trade-Off Analysis

Why Both O1 and O2 Are Essential

- **O1 alone (Priority Queue)**: Protects URLLC from packet drops, but during the 40s lag, mMTC queue backlog explodes (> 508 ms).
- **O2 alone (LSTM Autoscaler)**: Pre-warms containers and eliminates cold-start lag, but in a single FIFO queue, URLLC still suffers queue delays (333 ms).
- **O1 + O2 combined**: Provides both instantaneous fast-path delivery AND sufficient container capacity.

Infrastructure Cost Efficiency

- **Static Overprovisioning**: Keeps 6 AMFs running 24/7 $\Longrightarrow$ massive energy and cloud bill waste.
- **Warm Standby**: Burns 1024 AMF-seconds per surge.
- **Proposed LSTM Autoscaler**: Scales dynamically from 1 to 4 AMFs only when needed $\Longrightarrow$ **21.2% resource reduction**.

Conclusion

Summary of Completed Work:

1. **Resolved ScaleLB's Critical Weaknesses**: Solved the 40-second container startup lag and single-queue slice starvation.
2. **Guaranteed 3GPP SLA for URLLC**: Bounded URLLC 99th-percentile control latency to **3.15 ms** (< 5.0 ms SLA requirement) under a $10\times$ IoT flash crowd.
3. **Proactive Container Dimensioning**: Evaluated the LSTM lookahead rule ($H = 45$ s) ensuring zero cold-start overload windows.
4. **Verified Across Standard 5G Traces**: Validated across Flash Crowd, Diurnal Cycle, and Linear Ramp workloads.

Future Work:

- Integrating User Plane Function (UPF) slicing and session anchors.
- Deploying as an O-RAN Near-RT RIC xApp for edge microservice orchestration.

Thank You!

Questions & Discussion

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